

Freight Rate Prediction — Machine Learning Engineer Assessment

Final model: **CatBoostRegressor predicting rate-per-mile (RPM)**. The predicted RPM is multiplied by distance to produce the final freight rate.

1. Data and objective

The labeled development data contains 48,000 loads from January through October 2025. The assessment requires predictions for every row in the 12,000-row validation set and predictions for the 31 fixed December chart inputs.

2. Data quality and preprocessing

Negative weight values were treated as invalid and converted to missing. Missing/invalid weights were median-imputed using valid development weights, while missing/invalid indicators were retained. Categorical missing values were represented as UNKNOWN. The final model excludes `market_index` and `quote_signal` because they are not present in the fixed December scenario.

3. Feature engineering

Features include pickup, delivery, equipment, route, route plus equipment, distance transformations, weight, weight-per-mile, distance-weight interaction, geographic distance, coordinate deltas, calendar variables and cyclical date features.

4. Validation and model selection

A chronological split was used: January–September 2025 for training and October 2025 as the holdout. This avoids random temporal mixing and better represents future-period prediction. The final CatBoost RPM model used depth 5, learning rate 0.05, L2 regularization 5, seed 42, and the selected 170 iterations.

Metric	October holdout
MAE	\$102.79
RMSE	\$646.89
R ²	0.8209
Selected iterations	170

5. Final predictions

After selecting the iteration count using the October holdout, the model was refit on all labeled development rows. It generated predictions for all 12,000 validation loads and all 31 December rows. The supplied scorer validated both outputs successfully.

6. Fixed December prediction chart

The chart below is the chart generated by the supplied `score.py` from the completed December prediction file.

Candidate: December 2025 Predicted Load Rate

